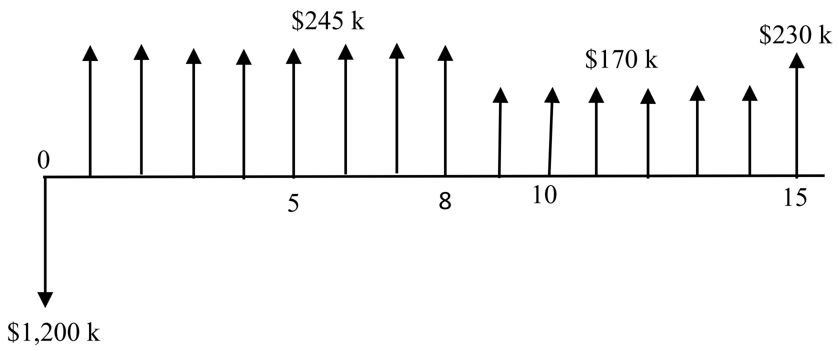
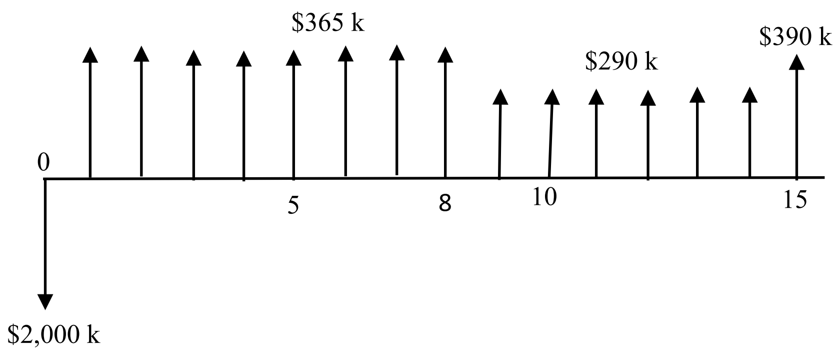
DSS 5202 Sustainable Systems Analysis
Solutions to Assignment #2

Question 1 (Cash flow diagrams)

Cash flow diagram for System A



Cash flow diagram for System B.



Question 2 (*NPV* method)

$$NPV_A(10\%) = \left[-1{,}200 + \sum_{k=1}^{8} \frac{245}{(1+0.1)^k} + \sum_{k=9}^{15} \frac{170}{(1+0.1)^k} + \frac{60}{(1+0.1)^{15}} \right] \times 10^3 \backslash$$

$$= \$\ 507{,}516.50 > 0$$

$$NPV_B(10\%) = \left[-2{,}000 + \sum_{k=1}^{8} \frac{365}{(1+0.1)^k} + \sum_{k=9}^{15} \frac{290}{(1+0.1)^k} + \frac{100}{(1+0.1)^{15}} \right] \times 10^3$$

$$= \$\ 629{,}821.73 > 0$$

$NPV_B(10\%) > NPV_A(10\%)$

Hence, System B should be selected based on the NPV method.

Question 3 (*IRR* method)

The incremental analysis method must be used for mutually exclusive alternatives using *IRR*.

IRR of System A is r such that

$$-1{,}200+\sum_{k=1}^{8}\frac{245}{(1+r)^k}+\sum_{k=9}^{15}\frac{170}{(1+r)^k}+\frac{60}{(1+r)^{15}}=0$$

Using an equation solver: r = 17.52% > *MARR* = 10%
System A is feasible.

IRR of System B is s r such that

$$-2{,}000+\sum_{k=1}^{8}\frac{655}{(1+r)^k}+\sum_{k=9}^{15}\frac{290}{(1+r)^k}+\frac{100}{(1+r)^{15}}=0$$

Using an equation solver: r = 15.48% > *MARR* = 10%
System B is feasible.

Base alternative with lower initial cost = System A

Next alternative = System B

Consider incremental cash flows (B – A)

IRR of (B-A) is r such that

$$-800+\sum_{k=1}^{8}\frac{120}{(1+r)^k}+\sum_{k=9}^{15}\frac{120}{(1+r)^k}+\frac{40}{(1+r)^{15}}=0$$

Using an equation solver: r = 12.57% > *MARR* = 10%
(B – A) is feasible.

Hence, System B should be selected based on the *IRR* method.

Question 4

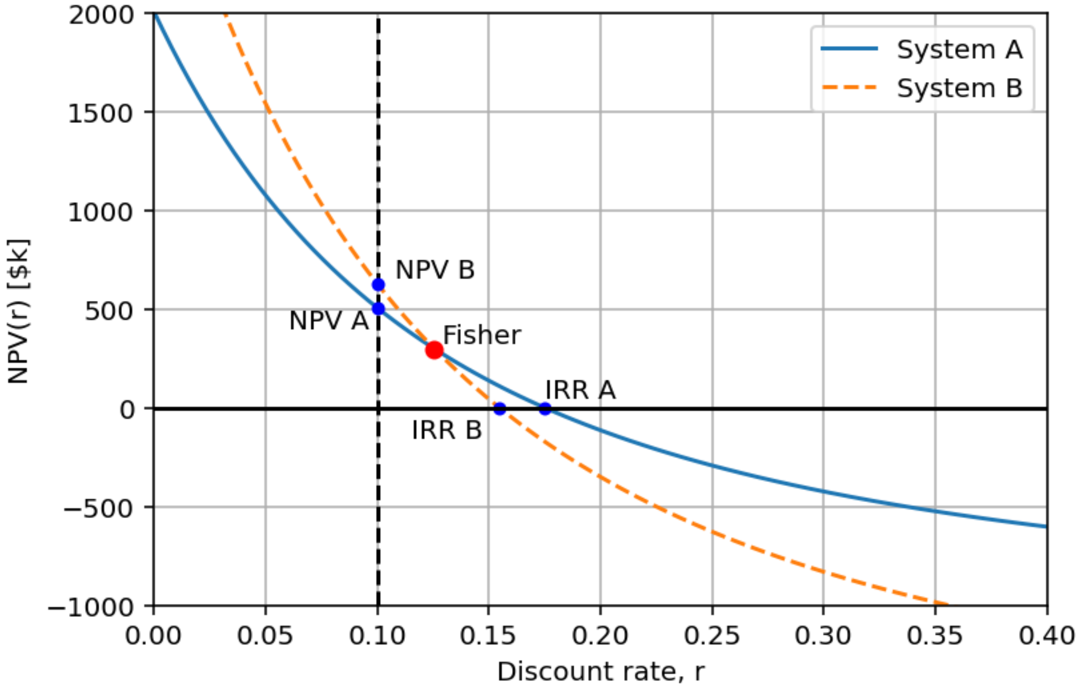
Comparison of NPV(10%) and *IRR* of the two alternatives.

	System A	System B
NPV(marr)	$ 507,516.50	**$ 629,821.73**
IRR	**17.52%**	15.48%

We do not expect the ranking of alternatives by *NPV*(*MARR*) to always be the same as the ranking by *IRR*.

The rankings will not be the same when the NPV(r) of the alternatives, plotted as a function of the discount rate r, intersect each other at the "Fisher intersection" that is greater than the *MARR*.

In our problem here, the Fisher intersection rate = 12.565%, and *MARR* = 10%.



Question 5 (*MIRR* of System B)

Financing rate = 8%
Reinvestment rate = 10%
MARR = 10%

Negative net cash flows occur in year 0 only:

| *PV*(-*ve* net CF at 8%) |
= - $ 2,000,000

Positive net cash flows occur in years 1 to 15:

FV(+*ve net* CF at 10%)

$$= \left[\sum_{k=1}^{8} 365\,(1+0.1)^{15-k} + \sum_{k=9}^{15} 290\,(1+0.1)^{15-k} + 100 \right] \times 10^3$$

= $10,985,417.99

$$MIRR_B = \sqrt[25]{\frac{10{,}985{,}417.99}{2{,}000{,}000}} - 1$$

= 12.026% > *MARR* = 10%

Hence, System B is financially feasible based on the *MIRR* method.